

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

To find the required current and bulb resistance :

- The battery's open-circuit voltage is 12 volts . The load dissipating 40 W across the battery terminals draws that power directly from the battery .
- Use $P = V^2 / R$ (for a fixed V), so $R = V^2 / P = 12^2 / 40 = 144 / 40 = 3.6$ ohms .
- The current drawn is then $I = P / V = 40 / 12 \hat{=} 3.33$ amps .

Therefore , the correct answer is E . 3 . 33 amps , 3 . 6 ohms .

</think>

<answer>

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</answer>

